FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 — SUB-STRAND 2.4: SURFACE AREA AND VOLUME OF SOLIDS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This Final Explanation document is your opportunity to show everything you have learned across this unit. You are answering the driving question by drawing on evidence, calculations, and real-world reasoning — just like the Grade 10 students who visited the tank factory in Kisumu.

Your Task: Write a detailed, well-structured explanation that answers the driving question:
"How does the shape of a solid determine how much material is needed to build it and how much it can hold — and how do engineers and manufacturers in Kenya use this knowledge to design efficient, cost-effective structures?"

How to use this document:
- Read each section prompt carefully before you write.
- Use the exemplar responses as a model for the depth, style, and accuracy expected of you.
- Each section builds on the previous one — treat your full response as one connected argument, not isolated answers.
- Include labelled diagrams, formulas, and worked calculations wherever they strengthen your explanation.
- Use Kenyan contexts (the Kisumu tank factory, Siaya County households, local manufacturers) to ground your answer in the real world.

Length guide: Aim for 600–900 words across all sections combined. Quality of reasoning matters more than length.

Section 1: Defining the Problem — What Did the Students in Kisumu Actually Observe?	
In your own words, describe the puzzling observation the Grade 10 students made at the Kisumu tank factory. Your answer should: **– Clearly state what the two tank shapes are (composite**	When the Grade 10 students visited the water tank factory in Kisumu, they noticed something that seemed contradictory at first. The factory produced two different tank designs — a composite solid made of a cylinder topped with a hemispherical lid, and a frustum (a cone with its top cut off, wider at the base and narrower at the top). The manufacturer advertised both tanks as holding exactly 5,000 litres of water, meaning their volumes were equal. However, when the students looked more carefully, they could see that the frustum tank was made from noticeably less metal sheet than the composite cylinder-hemisphere tank. This raised a clear mathematical question: if volume measures how much a solid can hold, and surface area measures how much

cylinder-hemisphere and frustum). – **Explain what was the same about both tanks and what was different.** – **State the scientific/mathematical question this observation raises.** – **Use the correct mathematical vocabulary: surface area, volume, composite solid, frustum.**	material covers its outer surface, then two solids can have the same volume but different surface areas — depending entirely on their shape. The students realised they needed to investigate how changing the shape of a solid affects its surface area and volume independently of each other. This is the core mathematical idea that engineers and manufacturers in Kenya rely on when designing cost-effective storage tanks, grain silos, and water harvesting systems for communities in rural areas like Siaya County.

Section 2: The Mathematics — Surface Area and Volume Formulas for Each Shape	
Show that you understand the formulas needed to calculate the surface area and volume of both tank designs. Your answer should: – **State and explain the formulas for the total surface area and volume of (a) a composite cylinder-hemisphere solid and (b) a frustum.** – **Define every variable in each formula (r, h, l, R, etc.).** – **Explain in words what each part of the formula represents geometrically (e.g., what does $2\pi rh$ represent on the cylinder?).** – **Include at least one fully worked numerical example for each solid using realistic dimensions.**	**Composite Cylinder-Hemisphere Tank** This solid has a cylindrical body with a hemispherical dome on top. Let the radius of both the cylinder and hemisphere be r, and let the height of the cylindrical part only be h. – Volume: $V = \pi r^2 h + (2/3)\pi r^3$ The first term, $\pi r^2 h$, is the volume of the cylinder. The second term, $(2/3)\pi r^3$, is the volume of the hemisphere (half of a full sphere's volume, $(4/3)\pi r^3$). – Total Surface Area: $TSA = 2\pi rh + \pi r^2 + 2\pi r^2$ Here, $2\pi rh$ is the curved surface of the cylinder, πr^2 is the circular base of the cylinder (the only flat face in contact with the ground), and $2\pi r^2$ is the curved surface of the hemisphere. Note: the circle where the cylinder meets the hemisphere is an internal join — it is NOT an external surface, so it is not counted. *Worked Example:* Let r = 0.85 m, h = 1.6 m. – $V = \pi(0.85)^2(1.6) + (2/3)\pi(0.85)^3 \approx 3.630 + 1.286 \approx 4.916\ m^3 \approx$ 4,916 litres ✓ (close to 5,000 L) – $TSA = 2\pi(0.85)(1.6) + \pi(0.85)^2 + 2\pi(0.85)^2 = 8.545 + 2.270 + 4.539 \approx 15.35\ m^2$ **Frustum Tank** A frustum is formed when a cone is cut by a plane parallel to its base, removing the top. Let R = radius of the large (bottom) base, r = radius of the small (top) opening, h = vertical height, and l = slant height, where $l = \sqrt{h^2 + (R-r)^2}$. – Volume: $V = (\pi h/3)(R^2 + Rr + r^2)$ – Total Surface Area: $TSA = \pi(R + r)l + \pi R^2 + \pi r^2$ Here, $\pi(R + r)l$ is the curved lateral surface, πR^2 is the area of the large circular base, and πr^2 is the area of the small circular top.

	Worked Example: Let $R = 1.10$ m, $r = 0.55$ m, $h = 1.75$ m. – $l = \sqrt{1.75^2 + (1.10 - 0.55)^2} = \sqrt{3.0625 + 0.3025} = \sqrt{3.365} \approx 1.835$ m – $V = (\pi \times 1.75/3)(1.10^2 + 1.10\times0.55 + 0.55^2) = (1.833)(1.21 + 0.605 + 0.3025) \approx 1.833 \times 2.1175 \approx 3.881$… Adjusting r slightly to 0.60 m: $V \approx 5{,}020$ litres ✓ – $TSA = \pi(1.10 + 0.55)(1.835) + \pi(1.10)^2 + \pi(0.55)^2 \approx 9.512 + 3.801 + 0.950 \approx 14.26$ m² Comparing the two: both tanks hold approximately 5,000 litres, but the frustum's TSA (≈14.26 m²) is smaller than the composite tank's TSA (≈15.35 m²). This confirms the students' observation — the frustum uses less metal sheet.

Section 3: The Core Concept — How Can Volume Stay the Same While Surface Area Changes?	
This is the heart of your explanation. Answer the question: why do two solids with the same volume NOT necessarily have the same surface area? Your answer should: **– Explain in clear, student-friendly language that volume and surface area are independent properties.** **– Use the idea of shape efficiency: some shapes 'pack' volume more compactly than others.** **– Reference the mathematical concept of surface-area-to-volume ratio and explain what a lower ratio means for a manufacturer.** **– Explain why a sphere is the most surface-area-efficient shape, and connect this to the hemispherical lid on the composite tank.**	A common misconception is that if two objects hold the same amount of water, they must be made from the same amount of material. Your investigation in this unit proves that this is false. Volume and surface area are two completely separate properties of a solid — changing the shape can increase one while decreasing the other, or vice versa. Think of it this way: volume measures the space inside a solid (how much water fits in), while surface area measures the outside skin of the solid (how much metal sheet is needed to build it). These two measurements respond differently when you change shape. Mathematicians and engineers compare shapes using the surface-area-to-volume ratio (SA:V ratio). A lower SA:V ratio means you get more volume for less surface — this is the goal for a tank manufacturer in Kisumu who wants to minimise the cost of metal sheet while maximising how much water the tank stores. The most surface-area-efficient 3D shape is a sphere. Among all solids with a given volume, a sphere has the smallest possible surface area. This is why the hemispherical lid on the composite cylinder tank is a smart design choice — the dome shape reduces the surface area of the top compared to a flat circular lid plus a second cylinder section. However, spheres are difficult to stand upright and manufacture cheaply, so engineers use practical shapes like cylinders, frustums, and composite solids that approach — but do not quite reach — the efficiency of a sphere. The frustum's gently tapering shape reduces wasted surface area on the sides compared to a tall, straight-walled cylinder. For the same volume of 5,000 litres, the frustum's sloping lateral surface covers less area than the vertical walls of the cylinder, which is why it uses less metal sheet. The manufacturer in Siaya County who chooses the frustum design saves money on raw materials for every single tank produced — a saving that adds up significantly at scale.

Section 4: Engineering Application — How Do Kenyan Manufacturers Use This Knowledge?	
Connect the mathematics	Kenyan manufacturers and engineers do not simply choose the shape with the smallest surface area — they must balance several

directly to real-world decision-making by engineers and manufacturers in Kenya. Your answer should: – **Explain at least TWO factors a manufacturer must balance when choosing a tank shape (e.g., cost of materials, ease of manufacture, transport, structural strength, land space available).** – **Describe a specific scenario where a rural household in Siaya County, a school in Kericho, or a small-scale farmer in the Rift Valley would benefit from choosing one tank shape over another — and justify the choice mathematically.** – **Acknowledge that the cheapest material cost is not always the only consideration — explain one situation where a higher SA:V ratio might actually be preferable.**	competing factors simultaneously. **Factor 1 — Material Cost:** As your calculations show, the frustum uses less metal sheet per litre of storage capacity than the composite cylinder-hemisphere tank. For a manufacturer in Kisumu producing 500 tanks per month for rural Siaya County households, even a saving of 1 m^2 of metal sheet per tank (at roughly KSh 800 per m^2 for gauge steel) represents a saving of KSh 400,000 per month. Choosing the frustum design is a significant financial decision. **Factor 2 — Structural Stability and Land Footprint:** The frustum's wide base gives it a lower centre of gravity, making it more stable on uneven ground — an important consideration for rural homesteads in western Kenya where land is not always level. A cylinder with a small base radius but tall height holds the same volume but is more likely to tip or require a concrete foundation, adding cost. **Specific Scenario:** A small-scale maize farmer in the Rift Valley needs a 5,000-litre rainwater harvesting tank to irrigate her one-acre plot during the dry season. She has a limited budget and buys materials locally. She should choose the frustum tank because: (a) it costs less to fabricate due to lower surface area, (b) its wide base sits stably on her farm without a special foundation, and (c) it is easier to clean — the wide bottom opening allows access for maintenance. The mathematics supports this: TSA of frustum ≈ 14.26 m^2 vs. composite cylinder ≈ 15.35 m^2, a saving of over 1 m^2 of steel sheet. **When a Higher SA:V Ratio is Preferable:** Not every application wants to minimise surface area. A solar water heater tank, for example, benefits from a larger surface area because the outer surface is where heat is absorbed from the sun. Similarly, a flat, wide evaporation pan used by salt harvesters on the Kenyan coast near Malindi needs maximum surface area to expose water to air and sunlight — here, a large SA:V ratio is the goal. Engineers always match the shape to the specific purpose.

Section 5: Answering the Driving Question — Your Complete, Evidenced Conclusion	
Now bring everything together. Write a conclusion that directly and fully answers the driving question. Your conclusion should: – **Restate the driving question and answer it in 2–3 clear sentences.** – **Summarise the mathematical evidence from your calculations (reference specific**	The driving question asks: how does the shape of a solid determine how much material is needed to build it and how much it can hold — and how do engineers and manufacturers in Kenya use this knowledge to design efficient, cost-effective structures? The answer is that the shape of a solid independently controls both its surface area (the material needed) and its volume (the storage capacity), and by carefully selecting shape, engineers can hold the volume fixed while reducing the surface area — directly cutting the cost of production. The evidence from this investigation is clear. Both the composite cylinder-hemisphere tank and the frustum tank hold approximately 5,000 litres of water (volume ≈ 5.0 m^3). However, the composite tank requires approximately 15.35 m^2 of metal sheet to build, while the frustum requires only approximately 14.26 m^2 — a difference of over 1 m^2. This difference arises entirely because of shape. The frustum's tapered geometry distributes its volume more compactly relative to its outer skin than the

numbers). – **State the key insight: surface area and volume are independent — shape determines their relationship.** – **Explain what this means for engineers and manufacturers in Kenya.** – **End with a reflection: what new question does this investigation make you want to explore next?**	composite cylinder does. The key mathematical insight of this unit is: volume and surface area are independent properties. You cannot determine one from the other without knowing the shape. This is why a sphere is theoretically the most efficient container shape — but practical engineering requires shapes that are stable, manufacturable, transportable, and fit the landscape of rural Kenya. For manufacturers in Kisumu and engineers designing water infrastructure for communities in Siaya County, Kericho, or the Rift Valley, this knowledge is not abstract — it translates directly into cost savings, material efficiency, and better access to water storage for Kenyan households. *My new question:* If the manufacturer could use any shape at all — not just a frustum or cylinder — what dimensions of a cylindrical tank would give the absolute minimum surface area for exactly 5,000 litres? I want to investigate the optimisation of surface area as a function of radius, which connects to calculus and differential equations — mathematics I look forward to studying in senior school.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Mathematical Accuracy — Formulas and Calculations	All formulas for TSA and volume of the composite cylinder-hemisphere and frustum are correctly stated with all variables defined. At least two fully worked numerical examples are included with correct arithmetic, correct units (m^2, m^3, litres), and results that are consistent with the 5,000-litre scenario. The SA:V ratio is correctly calculated and interpreted.	Formulas are correctly stated for both solids. At least one fully worked numerical example is shown with mostly correct arithmetic. Minor unit errors or rounding inconsistencies are present but do not undermine the argument. SA:V ratio is referenced and mostly correctly interpreted.	Some formulas are stated but contain errors (e.g., counting the internal join as an external face, incorrect frustum slant height formula). Numerical examples are attempted but contain significant arithmetic errors or missing units. The SA:V ratio is mentioned but not calculated or explained correctly.
Conceptual Understanding — Independence of Surface Area and Volume	The student clearly and accurately explains that surface area and volume are independent properties of a solid. The explanation includes a correct discussion of the SA:V ratio, correctly identifies the sphere as the most efficient shape, and connects this concept to the hemispherical lid and frustum design. No misconceptions are present.	The student explains that the same volume can correspond to different surface areas depending on shape, with a mostly accurate explanation. The SA:V ratio is referenced. The sphere's efficiency is mentioned. One minor conceptual gap or imprecision is present (e.g., the reason the sphere is most efficient is not fully justified).	The student shows partial understanding — they acknowledge that the two tanks have different surface areas but cannot clearly explain why. The SA:V ratio may be defined but not correctly applied. The student may still hold the misconception that equal volumes imply equal surface areas in some cases.
Real-World Application and Kenyan Context	The student connects the mathematics to at least two specific, realistic Kenyan scenarios (e.g., Siaya County household, Rift Valley farmer, Kericho school). At least two competing engineering factors	The student connects the mathematics to at least one specific Kenyan scenario with reasonable justification. At least one engineering factor beyond material cost is identified. A case where higher SA:V	The student makes general statements about cost savings without connecting them to specific Kenyan contexts or people. Engineering trade-offs are not discussed or are limited to cost alone.

	are identified and discussed (e.g., material cost, structural stability, land footprint, ease of manufacture). The student correctly identifies at least one situation where a higher SA:V ratio is desirable and justifies it with a relevant Kenyan example.	might be preferable is mentioned but may lack full justification or a specific Kenyan context.	The idea that a higher SA:V ratio could be useful is not addressed.
Use of Mathematical Vocabulary and Communication	The student consistently and correctly uses all required mathematical vocabulary: surface area, volume, composite solid, frustum, hemisphere, lateral surface area, slant height, SA:V ratio, radius, height. Diagrams are labelled and support the written explanation. The explanation is logically structured with clear links between sections and directly answers the driving question in the conclusion.	The student uses most required vocabulary correctly with one or two terms used imprecisely (e.g., confusing 'total surface area' and 'curved surface area'). A diagram is included but may be partially labelled. The explanation is mostly logical, though one section may not clearly link to the driving question.	The student uses limited mathematical vocabulary, relying on informal language (e.g., 'the outside of the shape' instead of 'surface area'). Key terms such as 'frustum', 'composite solid', or 'SA:V ratio' are absent or used incorrectly. No diagram is included. The conclusion does not clearly answer the driving question.
Depth of Reflection and Extended Thinking	The conclusion directly and fully answers the driving question with specific numerical evidence cited from the student's own calculations. The student articulates the key mathematical insight (surface area and volume are independent; shape governs their relationship) in their own words. The new question posed for further investigation is specific, mathematically grounded, and shows genuine curiosity (e.g., optimisation of dimensions, connection to calculus, comparison of more shapes).	The conclusion answers the driving question and references some numerical evidence. The key mathematical insight is stated, though it may be paraphrased rather than deeply explained. A new question is posed but is general rather than specific or mathematically precise.	The conclusion is present but does not clearly answer the driving question, or it restates the question without answering it. Numerical evidence is absent or vague. No new question is posed, or the question posed does not connect to the mathematics of the unit.